Subject: Mathematics | **PERIODIC TEST – 02 (PROBABILITY)** | **Time:** 60 Minutes
Class: Class X (CBSE) | **Chapter 14: Probability** | **MM: 30**

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Section A has 4 MCQs and 1 Assertion-Reason question (1 mark each).
2. Section B has 2 VSAQ questions (2 marks each). Section C has 2 SA questions (3 marks each).
3. Section D has 1 Long Answer HOTS question (5 marks). Section E has 1 Applied Case Study (4 marks).
4. Show clear steps in all probability derivations and calculations.

SECTION A: OBJECTIVE & MCQs (1 Mark Each) [5 Marks]

1. In a family of 3 children, the probability of having at least one boy is: **[1]**

Options: (a) $\frac{7}{8}$ (b) $\frac{1}{8}$ (c) $\frac{5}{8}$ (d) $\frac{3}{4}$

2. Two dice are thrown simultaneously. What is the probability of getting a sum greater than 9? **[1]**

Options: (a) $\frac{1}{6}$ (b) $\frac{5}{36}$ (c) $\frac{1}{9}$ (d) $\frac{1}{12}$

3. A card is drawn from a well-shuffled pack of 52 cards. The probability that the card is a spade or an ace is: **[1]**

Options: (a) $\frac{4}{13}$ (b) $\frac{17}{52}$ (c) $\frac{1}{4}$ (d) $\frac{2}{13}$

4. A bag contains 6 red, 4 white, and 8 blue balls. If three balls are drawn together, or if one ball is drawn, the probability that the drawn ball is not red is: **[1]**

Options: (a) $\frac{2}{3}$ (b) $\frac{1}{3}$ (c) $\frac{4}{9}$ (d) $\frac{5}{9}$

5. **Assertion (A):** If all red face cards are removed from a pack of 52 cards, the probability of drawing an ace from the remaining cards is $\frac{2}{23}$.
Reason (R): After removing 6 red face cards, 46 cards remain in the deck, including all 4 aces. **[1]**

(a) Both (A) and (R) are true, and (R) is the correct explanation of (A).
(b) Both (A) and (R) are true, but (R) is not the correct explanation of (A).
(c) (A) is true, but (R) is false.
(d) (A) is false, but (R) is true.

SECTION B: VERY SHORT ANSWER QUESTIONS (2 Marks Each) [4 Marks]

6. A die is thrown once. Find the probability of getting: (i) a number lying strictly between 2 and 6 (ii) an odd prime number. **[2]**

7. A lot consists of 144 ball pens of which 20 are defective and the others are good. A customer buys a pen selected at random. What is the probability that: (i) she will buy it (if it is good)? (ii) she will not buy it (if it is defective)? **[2]**

SECTION C: SHORT ANSWER QUESTIONS (3 Marks Each) [6 Marks]

8. Find the probability that: (i) a leap year has 53 Sundays and 53 Mondays. (ii) a leap year has 53 Sundays or 53 Mondays. (iii) a non-leap year has 53 Sundays. **[3]**

9. From the letters of the word 'MATHEMATICS', a letter is chosen at random. Find the probability that the letter chosen is: (i) a vowel (ii) the letter 'M' (iii) a consonant. **[3]**

SECTION D: LONG ANSWER / HOTS (5 Marks Each) [5 Marks]

10. From a pack of 52 playing cards, all face cards (Kings, Queens, Jacks) and aces are removed. The remaining cards are well shuffled and then one card is drawn at random. Find the probability of getting: (i) a black card (ii) a red card of value 8 (iii) a number card which is a multiple of 3 (iv) a card with a prime number (v) a card with an even number. **[5]**

SECTION E: APPLIED CASE STUDY (4 Marks) [4 Marks]

Question 11: Case Study Analysis

An environmental NGO surveyed 120 residents of a locality to study transportation choices made during their last vacation trip. The gathered data revealed:

- **Train:** 45 travellers
- **Personal Car:** 36 travellers
- **Aeroplane:** 24 travellers
- **Public Bus:** 15 travellers. A resident is chosen at random from this group.

Based on the above scenario, answer the following:

(i) What is the probability that the chosen resident travelled by car? **[1]**

(ii) What is the probability that the resident travelled by public mass transit (train or bus)? **[1]**

(iii) What is the probability that the resident did NOT travel by aeroplane? **[1]**

(iv) If this data is represented in a circular pie chart, what is the central angle corresponding to aeroplane travellers? **[1]**

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